

VR-Surv: a VR-Based Privacy Preserving Surveillance System

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ABSTRACT

The recent advances in smart city infrastructure have provided support for a higher adoption of surveillance cameras as a mainstream crime prevention measure. However, a consequent massive deployment raises concerns about privacy issues among citizens. In this paper, we present VR-Surv, a VR-based privacy aware surveillance system for large scale urban environments. Our concept is based on conveying the semantics of the scene uniquely, without revealing the identity of the individuals or the contextual details that might violate the privacy of the entities present in the surveillance area. For this, we create a virtual replica of the areas of interest, in real-time, through the combination of procedurally generated environments and markerless motion capture models. The results of our preliminary evaluation revealed that our system successfully conceals privacy-sensitive data, while preserving the semantics of the scene. Furthermore, participants in our user study expressed higher acceptance to being surveilled through the proposed system.

CCS CONCEPTS

• **Human-centered computing** → **Ambient intelligence**; • **Security and privacy** → **Privacy protections**.

KEYWORDS

Surveillance, Virtual Reality, Privacy, Procedural Generation, Motion Capture, Urban Environment

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1 INTRODUCTION

The recent advances in virtual reality have paved the way for a plethora of new applications to reach the big public. With the emergence of performant and relatively affordable hardware and software, the adoption of virtual reality has touched various sectors, like tourism [2], education, and emergency simulations and trainings [1]. The rising interest in virtual reality-based applications is justified by the flexibility that a virtual environment (VE) offers to

both, the designer and the user of the application. This flexibility is expressed in two major aspects. First, a VE constitutes a controllable environment, in which the content can be easily adapted to respect different criteria and constraints depending on the needs and ethics of the use case. Second, sensorial inputs from the real world can be used to augment the VE. Consequently, a VE benefits from what both worlds have to offer: the complexity of the real and the flexibility of the virtual. Therefore, VEs can be of great utility in bringing together controversial topics like video surveillance and privacy preservation. Achieving a fair trade-off between privacy and security presents considerable challenges [27]. The ubiquity of high-quality cameras and connectivity infrastructure have rendered video surveillance of large urban areas accessible to governments, companies, and individuals. With around 770 million CCTV cameras used around the world, video surveillance systems generate a considerable amount of information about citizens, sometimes without their knowledge or consent [28]. State-of-the-art techniques in deep learning and computer vision have made it possible to extensively analyze the content of surveillance videos. Object detection, emotion prediction, and activity recognition [3, 18] are now possible thanks to those models. Although necessary in several cases like security and traffic monitoring [16, 31], video surveillance raises several concerns about privacy. People's identities and behavioral patterns are exposed when recorded with traditional surveillance systems.

In this paper, we aim to leverage the capabilities offered by virtual environments to achieve an acceptable compromise between surveillance and privacy. We propose an immersive surveillance system for large-scale urban environments, that only conveys the semantics of real-world scenes, without revealing privacy-sensitive information. Our concept is based on creating a mapping between the real-world environment and a virtual version of itself. However, we constrain the virtual environment to flexibly obey rules covering two privacy-sensitive attributes: (1) the identity of the human subjects present in the scene, and (2) information about the spatio-temporal context of the recorded data. We achieve the former by using a neural network for pose estimation, to extract and track the poses of the human subjects that are detected in the scene. Then, we perform a real-time mapping of the tracked poses to rigged humanoid models in the virtual environment. For the contextual spatio-temporal information, we use satellite imagery and procedural content generation to recreate the identical environment layout while altering the appearance and the type of the buildings. A preliminary evaluation of the system compares its output to two representation forms: The traditional 2D video feed from CCTV and a point cloud representation. Hereby, we investigate whether participants can detect and perceive the content as good as in the traditional way. The results of our pilot user study (n=6) reveal that the quality of information conveyed by our system is comparable

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to the 2D video representation and is significantly better than the point cloud representation. Furthermore, users expressed interest in the immersive aspect and more acceptance of being recorded through the proposed system.

2 RELATED WORK

In this section, we go through the related works that shaped the design of our system. First, we address the most important privacy-preserving surveillance systems. Second, we focus on procedural environment generation, and markerless motion capture: Two techniques that represent the basis of our work.

2.1 Privacy Preserving Surveillance

Existing solutions toward privacy-preserving surveillance aim at protecting individual privacy-sensitive attributes. These attributes are personal markers that reveal biometric or behavioral information that can be used to identify individuals. To dissimulate the identity of the human subjects in a video surveillance feed, several solutions explore the concept of anonymized videos. These approaches are based on the intentional use of extremely low-resolution images that only allow for activity analysis or event detection. Instead of focusing on the details of the face, Miyazaki et al. [22] considered a two-steps approach, where they first rely on the shape of the head as a feature to detect persons in low-resolution surveillance videos. The detections are afterward confirmed by checking the shape of the entire upper body. In [6, 36], the authors have resorted to traditional computer vision operations like superpixel clustering, blurring, and mean shift filtering. Pei Li et al. [19] make use of a GAN-like generative neural network to learn a common feature space between the high- and low-resolution frames. Similarly, Zangnech et al. [38] learn a cross representation between faces in high and low-resolution images. This is done using a deep convolutional neural network with two branches to learn a non-linear mapping between the high resolution and low-resolution space. Although the above-mentioned approaches are effective and computationally efficient (because of the use of low-resolution input), their effect can easily be neutralized. High-resolution information can be retrieved through super-resolution neural networks [8, 9, 29], which have the capacity to denoise images and considerably increase the quality of the frames. Other techniques rely on cryptographic solutions to locally alter information in the images [37]. The encryption targets privacy-sensitive parts in the image like faces [11, 37], or windows on buildings to cover private spaces [12]. Encryption-based approaches are however incapable to scale to several detection models and visual tasks. Petrov et al. [24] eliminate the need to manipulate visual input from the early processing stages. It uses categorical activity recognition models to detect and signalize anomalous events to the user. Despite its lightweight, the approach completely relies on the performance of the deep learning model, which is problematic in the case of an undetected emergency event (false negative). The previously addressed approaches suffer from at least one of two major limitations: (1) They either sacrifice the quality of the input, sometimes inefficiently. (2) Or rely on prediction models (machine learning-based, or handcrafted) that rather consider simplistic semantic cues, which puts constraints on their detection capabilities (e.g.: makes them unable to detect

exotic emergency situations). Performing surveillance in a virtual environment represents an interesting alternative to overcome the limitations of traditional surveillance systems. If it is possible to reproduce motion, gestures, and the spatio-temporal context to a certain degree of fidelity in real-time, in a virtual environment, then altering privacy-sensitive attributes would not cause problems (using avatars instead of identical face attributes for example). Only few works have tried to approach surveillance from a virtual reality perspective. Early attempts were proposed in [26, 32]. The authors have implemented a virtual surveillance system that monitors a virtual environment. The environment was populated with AI-controlled agents and the goal was to provide a virtual environment to implement, test, and evaluate multi-camera surveillance systems, before deployment. The work in [25] built on top of [32] to simulate a smart camera network in a virtual reality-based replica of the Pen train station. In [15] authors used virtual reality to perform video surveillance of tourist attraction sites. However, privacy was not the major concern of those works. Furthermore, human agents in the scene were either generated by AI for simulation purposes or coarsely rendered based on GPS data. In this work, we aim to map the exact semantics and situations from the real environment to the virtual one, while keeping the place and the identity of the human subjects anonymous. Our work is based on recent techniques of procedural environment generation and markerless motion capture.

2.2 Procedural Environment Generation

Procedural content generation (PCG) is a generative technique to generate virtual environments based on procedural programming. Unlike the manual content generation that requires substantial efforts, the application of procedural content generation to the urban context allows for fast prototyping of large-scale virtual urban environments based on predetermined rule sets and grammars [30]. L-systems [20] in combination with procedural content generation are used in CityEngine [35] to create realistic 3D models for real cities. The generation is based on the overlapping of geographical data with statistics about behavioral motion patterns. Random3dCity [4] generates grid-based 3D urban scenes with n buildings, where n is dictated by the user. With the emergence of deep learning, several techniques leveraged deep neural networks for the generation process. Semantic segmentation models are used to parse scene components from real images and generate a level of detail (LOD) 1 model of the buildings in CityGML format [13]. As a result, several real-world cities models are generated for immersive applications like tourism [33], and urban planning [34]. In [10] a 3D model of the entire ancient city of Rome is reconstructed. In this work, procedural generation is however used as a privacy-preserving technique. We procedurally introduce changes to building shapes and textures, while keeping the same overall city look. The process is detailed in section 3.

2.3 Markerless Motion Capture

Motion capture is traditionally performed by installing reflective markers on the joints and surfaces of the subjects. Despite its precision, the technique remains constrained by the heavy setup and

complex sensors that it requires. With the advances in deep learning and computer vision, it is now possible to effectively perform motion capture without the need for sensors or particular acquisition environments. i.e. a markerless manner. Generally, the goal of markerless motion capture techniques is to provide a high-fidelity 3D motion reconstruction of individuals present in a scene, while, uniquely, relying on monocular or multi-view visual input. Mehta et al. [21] propose VNect, a deep learning-based framework that allows 3D human pose estimation in real-time. The model estimates the human body pose and fits a human skeleton model through frames to ensure temporal consistency. Despite its real-time capability and the possibility to use it in an outdoor environment, the model knows several limitations caused by depth estimation errors. Zhou et al. [39] apply weak supervision and make use of 2D labeled datasets. Their model is a two-stage network based on the hourglass architecture [23]. An example of temporal aspect consideration is found in [14], where the authors leverage the sequence-to-sequence architecture, essentially used in natural language processing, for pose estimation. The use of LSTM permits the inclusion of temporal context to robustly predict future poses. However, it limits the temporal window to 5 frames. To remedy the limited temporal window problem, [7] used temporal convolutional networks. Since our interest lays in pose estimation for multiple subjects in crowded urban scenes, we opt for a two-step framework-based model, called CrowdPose [17]. In such models, pose estimation is performed in two steps: First, the subjects of interest are detected. Then, human pose estimation is applied to the body proposals. Compared to other architectures, the two-staged approaches are more robust, since they build on top of the performant entity detection models.

3 CONCEPT

In this paper, we propose a VR-based video surveillance system, that preserves the privacy of individuals in urban areas, through real-time mapping of the real environment into a modified virtual version of itself. By introducing minimal changes to privacy-sensitive information in the scene, like faces and building facades, while retaining all the remaining information present in the scene, we only convey information about the semantics of the scene. Consequently, agents monitoring the environment through the proposed system would only evaluate the situation based on the activities, without identifying the subjects. The benefits of our proposed system are twofold: (1) protecting privacy-sensitive information, and (2) ensuring an objective situation assessment based on factual information extracted from the scene. We address the protection of two privacy-sensitive attributes. The first attribute is the identity and biometric information of individuals. The second attribute is the contextual information about the scene.

3.1 Identity Protection

To protect the identity of the individuals present in the real scene, we map their bodies to fully rigged humanoid models in the virtual environment. Every motion performed by the human subjects in the real scene is directly mapped in real-time to the virtual environment. Motion capture and mapping are performed as follows: For every frame coming from the camera, we perform joint person detection and pose estimation using the approach proposed in [17]. Only the

detected skeletons are forwarded to the virtual environment and mapped to the humanoid models. Furthermore, detected humans are tracked throughout the frames. The rendered virtual humanoids' positions and poses are updated and the temporal consistency of their motions is ensured.

3.2 Contextual Information Protection

Contextual information about the surveillance area, such as the exact place and time, plays a major role in detecting and analyzing behavioral patterns when combined with human detection and activity recognition [5]. To protect the contextual information while maintaining scene semantics, we use a bird-eye view satellite image of the area of interest along with publicly available meta-data about the roads and infrastructures and only extract the relevant scene layout, which serves as an input for the procedural scene generator. The output is a 3D scene model with the same layout and different building shapes and textures. The different scene representations are depicted in figure 1.

3.3 System Implementation

The system architecture is depicted in figure 2. In a first step, the virtual environment is created. For this, we delimit the area of interest and obtain its layout and related meta-data from OpenstreetMap¹. Then, we transfer the data to a Unity² scene, where buildings are procedurally generated. The procedural changes applied to shapes, textures, and types of the components composing the virtual scene are managed using Houdini³. For the live recording of the video data, we use stereo cameras that we place on real-world spots. The particular choice of stereo cameras is justified by their capability to provide depth information. In this way, we can ensure a better estimation of the relative positions of the detected individuals in the scene. The camera registration step is necessary to map the camera's properties and position to the virtual environment. After the system is set up, the live video feed of the cameras is processed in real-time to detect persons in the scene, perform pose estimation and forward the results to the skeleton mapping unit to render the results in the scene. As the area covered by the virtual environment is significantly larger than the area covered by the camera, the system can conceptually scale to multiple cameras that cover multiple locations. We propose two alternatives to interact with the system:

- An immersive version, where the navigation and the interaction with the scene are performed using HMD (in our case we used the HTC Vive⁴ HMD, controllers, and lighthouses.)
- A non-immersive variant in which the navigation and the interaction are done using a monitor and a mouse.

4 EVALUATION

As a first step to evaluate the proposed implementation, we investigate the capacity of the users to recognize human activities when using our system. We compare the output provided by our system to two baselines: (1) A video feed recorded using CCTV cameras,

¹OpenStreetMap: www.openstreetmap.org

²Unity: www.unity.com

³Houdini: www.sidefx.com

⁴HTC Vive: www.vive.com



Figure 1: The procedural generation of our virtual environment, depicted in (a), is based on a satellite image of the area of interest (b), and its meta-data in the form of map layout (c).

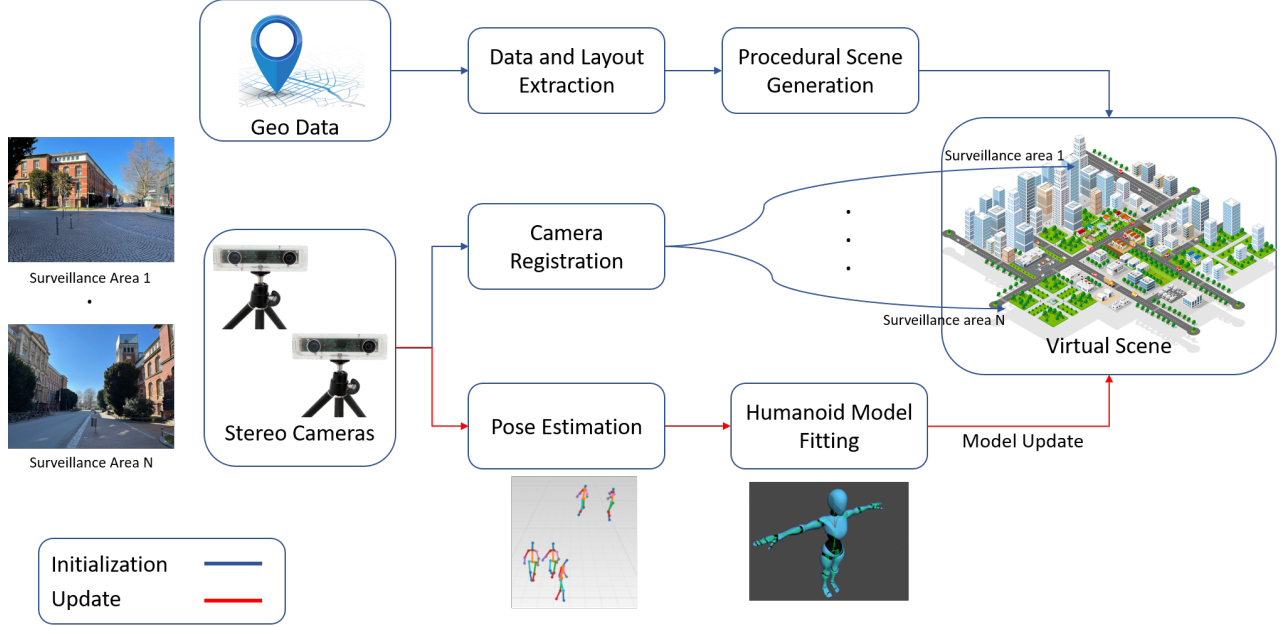


Figure 2: System Architecture of VR-Surv.

and (2) a 3D point cloud feed recorded using a LIDAR⁵, as depicted in figure 3. Whilst the first baseline represents the traditional and the most prominent way to perform video surveillance, the 3D point cloud recordings represent an interesting alternative for privacy preservation, as it is a 3D representation of the surrounding space that makes the recognition of the identities and the accurate spatio-temporal context challenging. We conduct a user study with 6 participants (4 male and 2 female) aged between 21 and 28 ($M=26$, $SD=2.7$) with an even former VR exposure. One participant owned a personal VR setup, and 2 other participants had at least one VR experience. The rest of the participants never tested a VR system. The participants were shown situations and activities performed by people in urban environments. Every participant was exposed to a total of 30 videos, 10 for each of the different visualization modalities, namely CCTV, 3D point cloud, and our system’s visualization. The length of the videos was between 10s and 15s. Specifically, we considered the 5 following situations: person walking, person running, person falling, people in conversation, people fighting. Moreover, we made sure that every situation is at least represented

⁵Light Detection and Ranging

once per visualization modality. The task assigned to the participants was to visualize the scenes and determine the situation in the shortest possible time. For every participant, we log the following metrics: (1) the average time per visualization modality, T_v , needed to classify the situation, calculated in milliseconds. (2) the number of correct recognitions per visualization modality, R_v . After the participants have visualized the 30 situations, a semi-structured interview was conducted to ask them about the quality of the representation, and how likely are they to accept being monitored by a similar system.

5 RESULTS

To analyze the results, A repeated-measures one-way ANOVA was used with the visualization modality as the unique independent variable. Mauchly’s test was used to test the sphericity assumption. For the recognition time, the sphericity assumption is violated. In that case, we use the Greenhouse-Geisser correction. We use the Bonferroni corrected pairwise t-test as a post-hoc measure, whenever we get significant results. In the following, we refer to

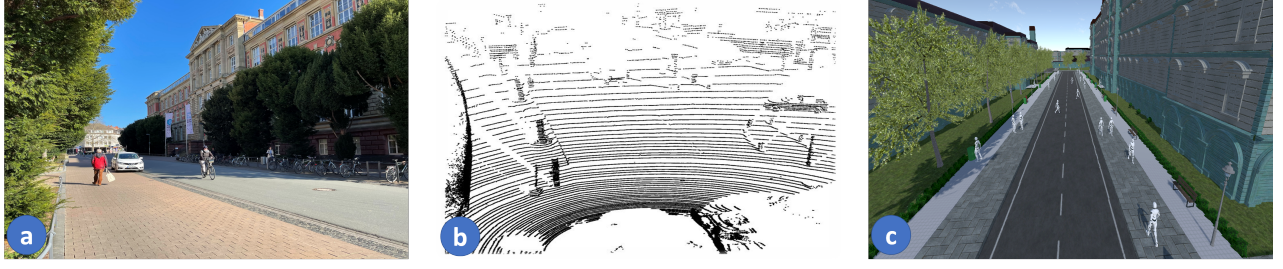


Figure 3: Modalities used in the user study for activity recognition: (a) CCTV, (b) point cloud scene representation, and (c) VR-Surv.

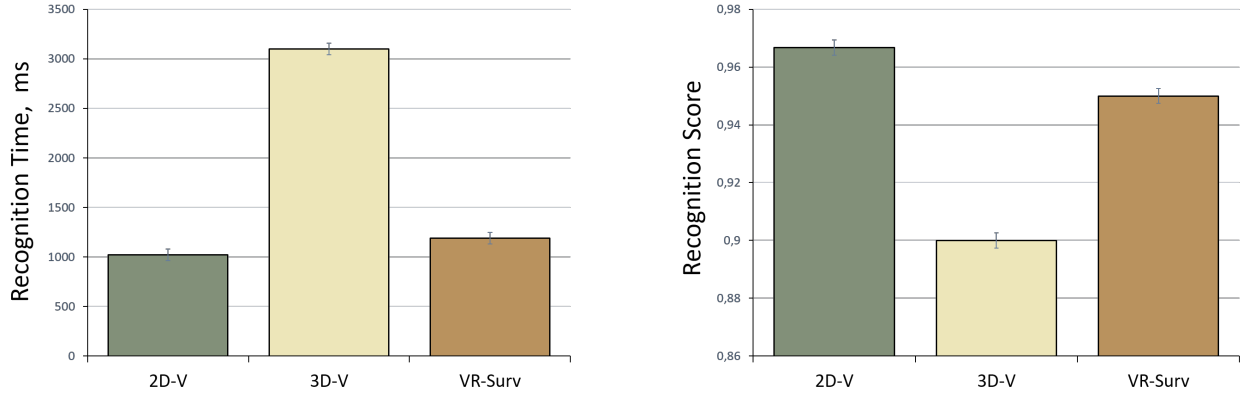


Figure 4: Overview of results: means and standard errors for the recognition time (left) and recognition score (right).

the CCTV-based and point cloud-based visualization modalities by 2D-V AND 3D-V respectively, to simplify the notation.

5.1 Recognition Time (T_R)

Considering the time it took the participants to recognize the presented situation, T_R , the 2D-V condition led to the fastest time ($M = 1021.33\text{ms}$, $SD = 116.01\text{ms}$), followed by the VR-Surv based visualization ($M = 1189.5\text{ms}$, $SD = 173.4\text{ms}$), and the 3D-V condition ($M = 3176\text{ms}$, $SD = 232.757\text{ms}$). A repeated-measures ANOVA revealed a significant difference between the conditions ($F(1.119, 5.594) = 299.686$, $p < .001$). The posthoc pairwise comparisons revealed a statistically significant difference between the 3D-V vs. 2D-V, and 3D-V vs. VR-Surv (all $p < 0.001$). The results are depicted in figure 4(left).

5.2 Recognition Score (S_R)

Considering the accuracy with which the participants recognized the presented scenarios, S_R , the participant achieved a comparable high performance for all the visualization modalities (2D-V : $M=0.9667$, $SD=0.0516$. 3D-V : $M=0.9$, $SD=0.08944$. VR-Surv : $M=0.95$, $SD=0.05477$). A repeated-measures ANOVA did not reveal any significant difference between the conditions ($F(2,10) = 1.226$, $p = 0.329$). The results are depicted in figure 4(right).

5.3 Qualitative Results

At the end of the experiment, the participants were asked to provide their feedback in the form of a semi-structured interview. All participants showed a strong interest in the idea of VR-based surveillance and confirmed a higher acceptance towards similar systems compared to the CCTV-based traditional approach. When asked about the quality of the scene representation and motion reconstruction, P1 stated that he could perfectly understand the different scenarios. P3 invoked the scalability potential of the proposed system and said that "it should be interesting to see how the system behaves with crowded scenes". From a privacy perspective, all participants were satisfied with the output. P5 highlighted the difference between VR-Surv and other systems. He stated that other systems would uniquely focus on simplistic identifiers like the face, whereas a combination between the identity (including clothes, sex, and other personal attributes) and context protection would offer a higher sense of privacy.

6 DISCUSSION AND FUTURE WORK

We have shown that our VR-based environment representation successfully provides an understandable and clear depiction of the events in real-time. Although participants achieved comparable performance in S_R by recognizing the activities with all the 3 proposed visualization modalities, it is to mention that the 3D point cloud-based representation is the variant that needed the most time.

The T_R values achieved by participants using our system are comparable to the CCTV-based baseline. We attribute this difference in time to the difference in clarity and information that each variant provides. The colored and detailed representation that our VR-based system provides, reduces the needed time to infer different activities and makes it as intuitive as watching a normal video. The immersive aspect of our system represented advantages compared to the normal video variant, as it allowed for better visualization of occluded zones in the 2D variant. The qualitative feedback of the participants was in favor of these observations. Participants showed a preference towards an immersive monitoring system and described it as more engaging compared to the other variants (P4 and P6). Furthermore, our system could convey a higher sense of privacy to the participants. Participants particularly liked that our representation did not only protect the identity of the individuals, but also targeted the entire context by using the procedural generation (P5). By neutralizing the appearance of the individuals and introducing changes to the spatial context in the scene, we assisted the participants in objectively evaluating the situations and reduced the personal subjective bias.

We see the present work as a starting point towards the application of VR in privacy preservation. In the future, a more detailed evaluation of the effect of different factors, like the level of detail and level of realism of the virtual environment, on the perception of quality and privacy is planned. Furthermore, we are interested in evaluating the system with more baselines, like low resolution representations, and with more challenging situations, like emergencies or disasters. Thereby, we target different potential stakeholders like firefighters, the police, or medical emergency teams.

7 CONCLUSION

We presented a design, implementation, and a preliminary evaluation of a VR-based immersive surveillance system. Our system provides a real-time, semantically rich, virtual reconstruction of a real-world surveillance zone, while at the same time protecting the individual's identities and the scene context. Our results showed that activities are easily recognizable using our system. Furthermore, its immersive aspect provides better coverage of occluded scene parts.

In a highly connected era where both privacy and security are equally needed, our system opens the path towards more acceptable, privacy-aware surveillance systems.

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